

1	2	3	4	5	6	sum
/20	/20	/20	/20	/20	/20	/100

Jan. 11 2014, Saturday

Name _____

Department _____

SID _____

University Physics

Final Examination

Kuang Yaming Honors School, Nanjing University

Some physical constants: $h = 6.63 \times 10^{-34} \text{ J} \cdot \text{s}$, $1 \text{ eV} = 1.60 \times 10^{-19} \text{ J}$, $m_e = 9.11 \times 10^{-31} \text{ kg} = 0.511 \text{ MeV}/c^2$,
 $1 \text{ u} = 1.660 \times 10^{-27} \text{ kg} = 931.5 \text{ MeV}/c^2$, $hc = 1.24 \times 10^3 \text{ eV} \cdot \text{nm}$.

Select five out of the following six problems.

1. A π^0 meson at rest decays into two photons ($\pi^0 \rightarrow \gamma + \gamma$).

(a) What type of interaction is mainly responsible for the decay?

(b) What are the energy and momentum of each photon?

(c) What is the wavelength λ of the photons? (The mass of the π^0 is $135 \text{ MeV}/c^2$.)

2. Consider a one-dimensional simple harmonic oscillator of mass m and natural circular frequency ω .

(a) Write down the Hamiltonian $\hat{H}$ of the oscillator.

(b) Show that the wave function $\psi(x) = Nx \exp\left(-\frac{\alpha^2 x^2}{2}\right)$, where $\alpha = \sqrt{\frac{m\omega}{\hbar}}$ and N is a constant, is an

eigenfunction of the Hamiltonian $\hat{H}$ of the oscillator.

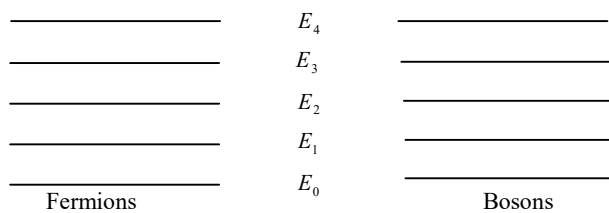
(c) Find $\langle x \rangle$, the mean value of the position of the particle, if the particle is in the state described by the wave function in (b).

3. (a) The energy levels a particle in the one-dimensional harmonic potential $V(x) = \frac{1}{2}m\omega^2 x^2$ can be given as

$E_n = (n + \frac{1}{2})\hbar\omega$, $n = 0, 1, 2, \dots$. If 8 identical fermions of spin 1/2 or 8 identical bosons of spin 0 are put in the harmonic potential, how the energy levels are filled in each case when the system is in the ground state? (Draw on the figure, $\uparrow$ as an up-spin fermion, $\downarrow$ as a down-spin fermion, $\circ$ as a boson.) What is the ground state energy of the system in each case?

(b) The Fermi energy E_F for the electrons in copper is measured to be 7.00 eV. If the mass of the electron is $m = 0.511 \text{ MeV}/c^2$, find the volume density n of the conducting electrons in copper.

Solution (a)



4. (a) An unstable element of half-life $T_{1/2}$ is produced in a nuclear reactor at a constant rate R . What is the equilibrium quantity of the element? How much time is required to produce 50% of the equilibrium quantity of the element?

(b) In terms of the rest mass of the parent atom M_p and the rest mass of the daughter atom M_D , determine the Q-values for β^- decay, β^+ decay and electron capture (K-shell capture):

$${}_Z^A\text{P} \rightarrow {}_{Z+1}^A\text{D} + e^- + \bar{\nu}_e \quad (\beta^- \text{ decay})$$

$${}_Z^A\text{P} \rightarrow {}_{Z-1}^A\text{D} + e^+ + \nu_e \quad (\beta^+ \text{ decay})$$

$${}_Z^A\text{P} + e^- \rightarrow {}_{Z-1}^A\text{D} + \nu_e \quad (\text{electron capture}).$$

5. The quark combinations of following particles are

$$\pi^- - d\bar{u} \quad \pi^+ - u\bar{d} \quad K^0 - d\bar{s} \quad K^+ - u\bar{s} \quad K^- - s\bar{u} \quad p - uud \quad \Xi^0 - uss \quad \Omega^- - sss.$$

(a) Find the charge number Q, the baryon number B and the strangeness S of these particles.

(b) Determine if the following reactions are mainly mediated by the strong or weak interaction, or are forbidden. Where a reaction is forbidden, give the reason.

(1) $K^- + p \rightarrow \Omega^- + K^+ + K^0$

(2) $\Omega^- \rightarrow \Xi^0 + \pi^0$

(3) $K^0 \rightarrow \pi^- + \mu^+ + \nu_\mu$

(4) $p \rightarrow e^+ + \gamma$

Quantum numbers of three quarks

Quark symbol	Charge number	Spin	Baryon number	Strangeness
u	2/3	1/2	1/3	0
d	-1/3	1/2	1/3	0
s	-1/3	1/2	1/3	-1

Solution: (a)

<i>Particle</i>	Q	B	S	<i>Particle</i>	Q	B	S
π^-				π^+			
K^0				K^+			
K^-				p			
Ξ^0				Ω^-			

6. Consider a spinless positive pion decaying at rest. (The initial angular momentum and linear momentum of the pion are zero.) A positive muon and a muon neutrino are emitted in opposite directions with equal and opposite momenta $\boldsymbol{p}_{\mu^+}$ and $\boldsymbol{p}_{\nu_\mu}$. Meanwhile, the two emitted particles have equal but opposite spins ($\boldsymbol{S}_{\mu^+}$ and $\boldsymbol{S}_{\nu_\mu}$) due to the angular momentum conservation.

- (a) Draw a figure to show the charge-conjugate process.
 (b) Can the charge-conjugate process occur in nature? Why?
 (c) What conclusion can you make if the charge-conjugate process can/cannot occur in nature?

